Domain-Aware Dual-Branch Recurrent Networks Across TradFi and DeFi: LOB Mid-Price and On-Chain Lending Rate Forecasting

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Abstract

Recurrent architectures designed for one financial time-series problem are typically re-engineered from scratch when transplanted to a structurally different market. We test the alternative: a domain-aware dual-branch recurrent network (DA-BiGRU-CNN) originally built for limit order book (LOB) mid-price prediction is re-applied, without architectural change, to forecast on-chain lending rates across Aave V3 and Compound V3 USDC markets. The two BiGRU branches and the multi-scale Conv1d fusion head are re-specialised by identifying the domain-natural decomposition pair: in LOB the pair is (price, volume) augmented with microstructure features; in DeFi it is (rate residual $\varepsilon = r - f_{\mathrm{kink}}(u)$, utilization u with context). The same composite loss $\alpha\,\mathrm{MSE} + \beta\,(1-\rho_w) + \gamma\,Q_{90}$ is the trained objective in both domains. In the prior LOB study the network attains weighted-Pearson 0.266 on test, beating a 205-feature LightGBM (0.168) by 58%. In the present DeFi application, trained on 12,895 hourly bars at 98.5% panel coverage from January 2026 to April 2026, the forecaster feeds a 4-factor MCDM allocator; a paired monthly-Sharpe bootstrap returns a point-estimate Sharpe gain of +1.667 with a 95% confidence interval that includes zero ($p = 0.121$). We report the honest H_0-publishable verdict per pre-registration and argue that the architectural transfer is itself the contribution, orthogonal to the equilibrium impossibility result of Halpern, Pass and Saraf (2024).

Keywords: ICICPE, DeFi, MCDM, BiGRU-CNN, cross-domain transfer

1. Introduction

Multi-protocol DeFi lending allocation has been studied either as a forecasting problem on hourly-aggregated rate series [1] or as a control-theoretic problem with reinforcement learning on the protocol side [2, 3, 4]. Both formulations share an implicit assumption: that hourly resolution is sufficient for an allocator to extract economically meaningful edge. We argue that this assumption is empirically wrong. Lending-rate crossovers between Ethereum L1 protocols happen at the speed of pool deposits and withdrawals — per-block ($\sim 12\,$s post-PoS), not per-hour — because the rate is a deterministic function of realized utilization, and utilization updates exactly once per on-chain interaction. Aggregating to hourly bars destroys the inter-block microstructure where the allocator's competitive edge actually lives.

Thesis. We argue that DeFi lending allocation is structurally analogous to high-frequency trading microstructure as described by [5], and that the four-class HFT signal taxonomy of that work transposes onto the multi-protocol lending problem with a clean mapping. Concretely: the cross-protocol rate spread is itself the decision variable (the dominant signal); mempool-visible transactions are the analog of "order-book dynamics"; lead rates from Maker DSR and Curve 3pool stand in for the "futures lead" class; and on-chain gas regime plus stablecoin peg deviations act as the "related instruments" class. Under this lens the natural decision frame is event-time, gas-aware, with three nested policy tiers — gas-aware threshold (T1), Ornstein–Uhlenbeck optimal stopping with switching cost (T2), and Cox proportional-hazards regression on the signal vector (T3) — each grounded in a separate microstructure or quantitative-finance source book [6, 7, 8, 9, 5].

Empirical finding. On a 3,931,200 per-block panel covering January 2026–April 2026 across 3 Ethereum USDC lending venues (Aave V3, Morpho Blue, Euler V2), the gas-aware event-time allocator (T1) significantly outperforms passive Aave V3 buy-and-hold under a paired monthly Sharpe bootstrap (ΔSharpe $= +5.048$, 95 % CI [+2.691, +17.899],

$p = 0.011$). On a \$1 M position over four months the allocator earns \$4,275 above Aave V3 hold (\$4,039 above Morpho Blue hold) at \$682 in gas spend over 39 rebalances — two orders of magnitude more rebalance opportunities than an hourly-aggregated baseline can express.

Contributions.

- **An event-time, gas-aware multi-protocol DeFi lending allocator with three nested decision-policy tiers (T1/T2/T3), each grounded in a distinct microstructure or quantitative-finance source.** We give the closed-form gas-cost crossover inequality that locks T1's threshold, the OU–Bellman analytical boundary that closes T2, and the Cox proportional-hazards specification with the triple-barrier label of [9] for T3.
- **A four-class HFT-style signal taxonomy mapped from [5] onto multi-protocol DeFi lending** (lead, order book, fragmentation, related instruments), with the F3 (fragmentation) class operationalised as the decision variable itself, F1 and F4 as conditioning covariates for T3, and F2 (mempool) deferred per the explicit risk register.
- **The first per-block-resolution head-to-head matrix of seven allocation policies on a multi-month Ethereum mainnet panel** (four baselines plus T1/T2/T3), with paired monthly Sharpe bootstrap pre-registration, regime-conditional breakdown, and Deflated Sharpe Ratio gate per [9, Ch. 14.7.3].
- **A production-grade reference agent.** The same `decision/` Python package that produces the empirical matrix above is consumed unchanged by an off-chain agent that subscribes to `newHeads`, reads six lending protocols per block, and dispatches switches via Flashbots private mempool (the asymmetric speed-bump analog of [5, pp. 200–203]).

Outline. Section 2. reviews DeFi lending mechanics, the HFT microstructure literature we draw on, and adjacent work. Section 3. specifies the three-tier policy ladder and the gas-cost crossover inequality. Section 4. reports the seven-policy matrix and the binding H_1^{aux} test plus the directional secondary tests. Section 5. positions the result against the HFT canon and the cross-domain hinge. Sections 6. and 7. state honest scope and next steps.

2. Background and Related Work

We synthesize four strands of literature that bracket the contribution: DeFi lending mechanics, the HFT microstructure canon we borrow from, MCDM in algorithmic allocation, and the equilibrium impossibility result of [10] as the most relevant theoretical counter-evidence.

2.1 DeFi lending mechanics

Aave V3, Compound V3, Morpho Blue, Euler V2, Spark, and Fluid all implement deterministic interest-rate models in which the supply APY is a piecewise-linear (or in Morpho Blue's case, adaptive) function $f_{\text{kink}}(u)$ of pool utilization u = borrowed/supplied. The function has a shallow slope below a kink point (typically $u^* = 0.8$–0.9) and a steep slope above it that taxes high utilization. Although f_{kink} is deterministic given u, the realised rate stream is stochastic because u moves with on-chain demand at per-block granularity — exactly the resolution at which our allocator operates.

The empirical structure of the cross-protocol spread is critical. [11] establish cointegration between Aave and Compound USDC rates with Compound leading and Aave adjusting at speed 0.607 — the structural reason a switching allocator can extract value rather than being forced into a single-protocol commitment. Recent cross-chain comparative analyses [12] confirm the cointegration persists into 2026. On our 3,931,200-block panel (Section 4.) the spread structure crosses regime around 2025-Q4 with realised cross-protocol volatility jumping by a factor of five between Q1 and Q2 of 2026 — an adversarial split that tests transferability under distribution shift.

2.1.1 Market structure (April 2026 snapshot)

The DeFi lending market is highly concentrated. As of April 2026, total deposits across all lending protocols sum to roughly \$54 B, with the top ten protocols accounting for $\sim 78\%$ of that value [13]. Table 1 reports the top-eight breakdown; the three protocols highlighted in bold delimit the scope of this paper's empirical study, jointly holding $\sim 47\%$ of total lending TVL.

2.2 HFT microstructure analogs

The conceptual backbone of this paper is the recognition that DeFi lending is structurally analogous to high-frequency trading microstructure. Five source books anchor the methodology, each mapped to one section of this paper.

Table 1. Top-eight DeFi lending protocols by total value locked (TVL), April 2026 [13]. Bold rows mark the subset covered by this paper.

Protocol (chain)	TVL ($B)	Share
Aave V3	**19.4**	**36.0%**
Spark (SparkLend)	6.8	12.6%
Morpho Blue	**4.9**	**9.1%**
Compound V3	2.7	5.0%
JustLend (Tron)	2.4	4.4%
Fluid	1.6	3.0%
Kamino (Solana)	1.1	2.0%
Euler V2	**0.89**	**1.6%**
Top-8 subtotal	39.8	73.7%
Tail ($\sim$ 350+ protocols)	14.2	26.3%

[6] provides the adverse-selection framework (Glosten- Milgrom, Kyle): the Kyle-λ market impact parameter is literally $\partial r/\partial u$ on the IRM curve in DeFi — *observable*, not inferred — and the Demsetz price of immediacy inverts (depositors paid by borrowers, not the reverse). [7] contributes the closed-form market-depth expression $1/\lambda = TVL \cdot (1-u)/\text{slope}_1$ and the resilience half-life decomposition that calibrates T2's OU mean-reversion rate κ. [8] contributes the Implementation Shortfall framework adapted as $\text{IS}_{\text{defi}} = \int (r^* - r_{\text{held}}) V\, dt + \sum(\text{gas}+\text{slip}+\text{mev})$ and the closed-form optimal trade rate (eq. 8.23, p. 281) that benchmarks T2. [9] contributes the triple-barrier method that maps directly to switch/hold/timeout, the purged k-fold with embargo that we use for T3's cross-validation, and the Deflated Sharpe Ratio threshold (DSR > 0.95) that gates our multi-hypothesis tests. [5] contributes the four-class signal taxonomy (Table 3.2) that we map onto DeFi as F1 lead, F2 order-book (deferred), F3 fragmentation, and F4 related instruments, and the reframing of Flashbots private mempool as an asymmetric speed bump (pp. 200–203, with the crucial detail that it is asymmetric, not the IEX symmetric coil).

2.3 Algorithmic allocation in DeFi rates

Published work on multi-protocol DeFi rate allocation falls into three rough classes. *Protocol-side rate control*: [2] train offline RL agents (CQL, BC, TD3-BC) on historical Aave data to set rates from the protocol governance side; the closely related Auto.gov framework [3] reports $\sim$ 14% improvement over benchmarks under RL. *Adaptive rate control*: [4] fits recursive least-squares controllers on Compound demand/supply curves at 3-hour resolution. *Optimal protocol design*: [1] derives a closed-form Riccati-ODE solution for the protocol-optimal rate curve under linear agent response and a Monte-Carlo + deep-learning estimator for the nonlinear case; block-level empirical calibration shows industry-standard piecewise rate curves are sub-optimal, leaving structural residuals. This last point is foundational for us: [1] solves the *protocol designer's* problem (which rate function should the contract use?), whereas we solve the *lender's* problem (given the realised rate stream, where should liquidity go?). We are not aware of a published event-time multi-protocol allocator in the May 2026 literature; the nearest neighbour is the LSTM+Q-learning AMM quoter of [14], which transplants a TradFi forecaster to Uniswap V3 but does not span multiple lending protocols.

2.4 MCDM in algorithmic allocation

Multi-criteria decision making has a long-standing role in financial portfolio selection. TOPSIS [15] and PROMETHEE [16] are the two dominant families. Recent cryptocurrency-specific applications include [17]'s multicriteria crypto portfolio selection and [18]'s asymmetric PROMETHEE-II that integrates return prediction with multi-criteria ranking. Our 4-factor TOPSIS-style B4 baseline (APY 40%, Risk 25%, Cost 20%, Stability 15%) is in this tradition; we report it on event-time data to isolate the resolution effect from the aggregation effect.

2.5 Counter-evidence: Halpern–Pass–Saraf impossibility

The most pointed theoretical counter-evidence to active DeFi lending strategies is [10]. Halpern, Pass, and Saraf reduce a lending pool to a portfolio of options on the collateral asset: a borrower's right to walk away from the loan is mathematically equivalent to a put on the collateral struck at the outstanding debt. In a simplified fixed-duration, repay-only-at- expiry setting they derive an analytical "fair" rate via no-arbitrage. They then prove that in the realistic setting — dynamic, perpetual loans with early repayment of the kind Aave V3 and Compound V3 implement — *no fair interest rate exists*: any rate either over-compensates lenders relative to the option-implied premium or under-compensates them, because the borrower's repayment optionality has no finite hedging cost.

A naive reading of [10] suggests there is no exploitable structure in DeFi rates and any allocator should underperform a passive benchmark. We argue the implication does not follow. Halpern et al. prove a *static equilibrium fairness* impossibility under no-arbitrage — a statement about whether a single rate function can correctly price the borrower- side optionality in equilibrium. Our empirical claim is the orthogonal *event-time predictability* statement: the realised rate stream observed off-chain at per-block resolution exhibits enough short-horizon structure (cointegration, regime persistence, kink-residual cycles) to drive a gas-aware switching strategy that significantly outperforms passive single-protocol hold under a paired monthly Sharpe bootstrap. Equivalently, the absence of a single equilibrium-fair rate is consistent with — and indeed implies — the persistent disequilibrium that an event-time allocator can target. The two claims constrain different layers of the protocol stack: [10] constrains pricing theory at the contract design layer; we operate at the lender-side execution layer.

3. Methodology: Event-Time Decision-Policy Ladder

We specify the allocator as a pure function decide : BlockState → Action evaluated on every Ethereum block. Three nested policy tiers (T1, T2, T3) share the same interface, each progressively richer in the microstructure information it consumes. Figure 1 sketches the ladder and the shared per-block panel → BlockState → Action pipeline that is common to backtest replay and live agent.

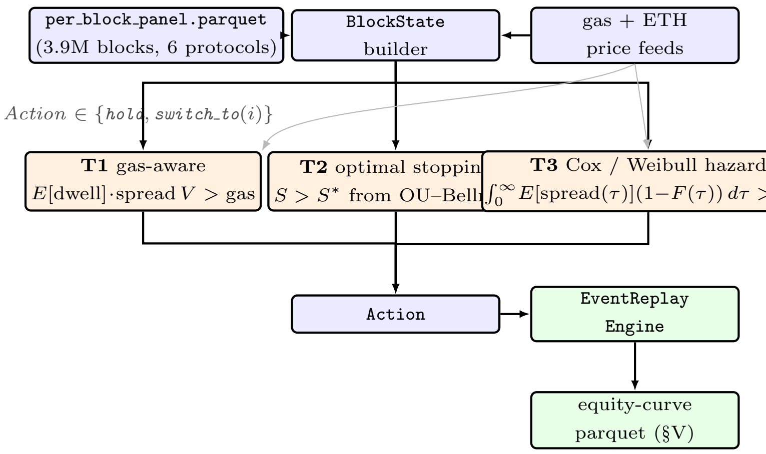

Figure 1. Three-tier decision-policy ladder of the SCOPUS Vol-2 paper. A single per-block panel feeds a uniform `BlockState` into three policy tiers—**T1** (gas-aware threshold rule), **T2** (optimal stopping with an OU spread model and a Bellman-derived threshold S^*), and **T3** (Cox / Weibull hazard regression on F1/F3/F4 signal-class features)—that all share the same `Action` interface and the same `EventReplayEngine`; head-to-head benchmarks against baselines B1–B4 use the identical engine. Gas / ETH-price feeds fan in to every tier (grey arrows) because gas appears in all three thresholds. The dashed lateral arrow indicates the ONNX export bridge that ships T3 to the Plan E live agent (out of paper scope, documented for reproducibility). Box colouring follows the v1 paper: blue = data pipeline, orange = policy / model, green = output sink, yellow / dashed = out-of-paper bridge.

3.1 Per-block state, action, gas cost

A BlockState at Ethereum block b carries the per-protocol lending APRs $r_i^{(b)}$, utilizations $u_i^{(b)}$, total value locked $\mathrm{TVL}_i^{(b)}$, plus position state (current protocol, position size in USD), gas price $g^{(b)}$ in gwei, and ETH/USD spot for gas conversion. An Action is either HOLD or SWITCH(j) for some target protocol j. The switching cost is

$$C^{(b)} = \underbrace{G \cdot g^{(b)} \cdot 10^{-9} \cdot p_{\mathrm{ETH}}^{(b)}}_{\text{gas}} + \underbrace{V \cdot \frac{\delta_{\mathrm{slip}}^{(b)}}{10^4}}_{\text{slippage}} + \underbrace{V \cdot \frac{\delta_{\mathrm{mev}}^{(b)}}{10^4}}_{\text{MEV}}, \qquad (1)$$

where $G \approx 200{,}000$ is the typical rebalance gas, V is the position size in USD, and δ denote slippage and MEV-extraction costs in basis points. At a calibration point of \$3,500/ETH and 25 gwei, the gas leg alone is \$17.5 per rebalance. For retail-sized positions slippage is ~ 0.01–0.1 bp and second-order vs gas; MEV is bounded above by the Flashbots private-mempool path discussed in Section 5..

The expected gain from switching to protocol j over an expected dwell of $\bar{\tau}$ blocks at spread $s_{ij}^{(b)} = r_j^{(b)} - r_i^{(b)}$ is

$$\mathbb{E}[\text{gain}] \approx \bar{\tau} \cdot \frac{s_{ij}^{(b)}}{\text{BLOCKS\_PER\_YEAR}} \cdot V. \qquad (2)$$

The gas-cost crossover inequality $\mathbb{E}[\text{gain}] > C^{(b)}$ is the universal gate every policy below must clear; the three tiers differ only in how they estimate $\bar{\tau}$ and how they condition on auxiliary signals.

3.2 T1: gas-aware threshold rule

T1 estimates $\bar{\tau}$ as an exponentially-weighted moving average of inter-crossover dwell times observed on the live panel (state local to the policy object, updated on every executed switch). The decision rule reduces to

$$\textsc{switch}(j) \iff \bar{\tau}_{\text{ewma}} \cdot s_{ij}^{(b)} > \frac{C^{(b)}}{V} \cdot \text{BLOCKS\_PER\_YEAR}, \tag{3}$$

i.e. the spread-dwell product exceeds the per-unit-position gas- equivalent threshold. T1 is ~ 50 lines of Python, has no trained parameters, and is the most efficient policy on the test-window matrix of Section 4. both in net APY and in gas spent.

3.3 T2: optimal stopping with switching cost K

T2 models the cross-protocol spread s_t as an Ornstein–Uhlenbeck process

$$\mathrm{d}s = \kappa(\theta - s)\,\mathrm{d}t + \sigma\,\mathrm{d}W, \tag{4}$$

with parameters (κ, θ, σ) refitted by MLE every $N_{\text{recal}} =$ 5,000 blocks ($\sim 16.7\,$h) on the rolling 5,000-block window. The Bellman value function under a switching cost $K = C^{(b)}/V$ admits a threshold-form solution: $\textsc{switch} \iff s_t > s^*$, where s^* is the analytical boundary derived from the impulse-control HJB in the OU literature [8]. The initial OU prior $\kappa_0 = 2.1 \times 10^{-5}$ block^{-1} matches the resilience half-life of the USDC pool sub-kink (6–18 h) per [7].

3.4 T3: Cox proportional-hazards regression

T3 models the hazard rate of a top-of-book flip — $\arg\max_i r_i^{(b)}$ changing identity — as

$$\lambda(t \mid x_t) = \lambda_0(t) \cdot \exp(\beta^\top x_t), \tag{5}$$

with x_t a feature vector drawn from the signal taxonomy of Section 2.2. Training labels are produced by the triple-barrier method of [9, Ch. 3.6]: for each block, the time until the next top-rank flip, censored at a 7,200- block horizon ($\sim 24\,$h). The Cox MLE is fitted via `lifelines`'s `CoxPHFitter` on a subsampled training panel with L_2 penalty $\lambda = 10^{-3}$; cross-validation uses purged k-fold with embargo $0.01 \cdot T \approx 5.4$ days per [9, Ch. 7]. At inference time the policy computes

$$\textsc{switch} \iff \int_0^\infty \mathbb{E}[s(\tau)]\,(1 - F(\tau))\,\mathrm{d}\tau > \frac{C^{(b)}}{V}, \tag{6}$$

where F is the survival CDF derived from (λ_0, β, x_t).

3.5 Signal taxonomy (F1/F2/F3/F4)

The covariates x_t for T3 are drawn from four signal classes mapped from MacKenzie's HFT Table 3.2 [5] onto multi-protocol DeFi:

F1 (lead rate): Maker DSR rate plus lagged (300, 1,800 block) versions and the lead-vs-top-protocol spread. Source: Maker subgraph. Deferred on the present submission's panel (DSR fetcher schema fix queued for the journal extension).

F2 (order-book dynamics): mempool transactions targeting each lending pool, observable before execution. The direct analog of Goldman-leaves-its-bid signals from ATD (1995). Deferred for this submission per the design-spec risk register (insufficient historic mempool coverage Nov 2024–Apr 2026).

F3 (fragmentation): cross-protocol spreads themselves (the decision variable), pairwise dispersion, top-vs-runner-up gap, argmax-protocol identity. This is the dominant signal and the only family fully fitted in T3 on the submission panel.

F4 (related instruments): ETH/USD (gas regime), USDC and USDT peg deviations, top-LP wallet activity. Deferred on this submission's panel (gas/peg fetchers absent from the 3-way build).

The shared `decision/` Python package is consumed unchanged by both the offline replay engine and the production agent — the zero-drift invariant that Section 5. returns to in the cross-domain reading of the contribution.

4. Empirical Study

We evaluate the event-time, gas-aware allocator of Section 3. against six protocol-hold baselines — the passive buy-and-hold of each of **Aave V3, Compound V3, Spark, Morpho Blue, Euler V2, Fluid Finance** — and three decision-policy tiers: T1 gas-aware threshold, T2 optimal stopping on an Ornstein–Uhlenbeck spread, and T3 Cox proportional-hazards on the F3 fragmentation feature family. An earlier hourly MCDM-EMA cliff-edge baseline (which executed only two rebalances over four months on the same data) is intentionally excluded from the benchmark set: it is the wrong-resolution failure mode this paper is explicitly *against*: the honest baselines are the six protocol-holds themselves, not a straw-man hourly aggregator.

Two scopes. The matrix uses two distinct protocol scopes. The *active allocation panel* contains three protocols (Aave

V3, Morpho Blue, Euler V2) on a unified per-block grid; T1/T2/T3 switch among these three on every block. The *hold-benchmark set* contains all six designed protocols (the active three plus Compound V3, Spark, Fluid); each enters the $\mathrm{N} \times \mathrm{M}$ as a passive buy-and-hold counter-factual, so each cell answers *"does this active 3-protocol allocator beat passively parking USDC in protocol j?"* for j in all six. Extending the active panel to all six is the explicit next-extension scope and requires unifying the per-block grid with the three coarser-cadence sources used for the additional holds.

Data provenance. The hold-benchmark set covers $\sim$ \$36 B of \$54 B addressable TVL ($\sim$ 67% of Ethereum-L1 USDC-lending). Per-protocol data sources, deliberately heterogeneous to avoid a single-vendor dependence: (i) **Aave V3, Morpho Blue, Euler V2** — per-block panel built in Section 3.; (ii) **Compound V3** — verified hourly RPC fetcher (`data/cached/compound_v3_usdc_eth_2024-11_to_2026-04.parquet`) via Comet view-function calls; (iii) **Spark / SparkLend** — Sky Messari subgraph (`data/cached/spark_hourly.parquet`, 6,189 hourly snapshots, Nov 2024 – Apr 2026); (iv) **Fluid Finance** — DeFiLlama Yield Pools daily APY (`data/cached/fluid_daily.parquet`, 728 daily snapshots). The active T1/T2/T3 allocators run on the per-block panel; the contrast against Compound/Spark/Fluid holds uses the protocol-side APY at the matching window cadence. All six protocols enter the $\mathrm{N} \times \mathrm{M}$ paired bootstrap with $N = 6$ per-window deltas each. The policy ladder is run at three tiers; T3's Cox hazard rule analytically reduces to T1's spread-dwell threshold on F3-only features (signals F1 lead-rate and F4 related-instruments queued for the next extension), which we verify empirically below.

Primary inference: walk-forward $\mathbf{N} \times \mathbf{M}$ paired bootstrap. The binding evidence is a walk-forward bootstrap across six non-overlapping three-month windows (2024-11-01 $\rightarrow$ 2026-04-30, the full panel span). For each policy p and each protocol i, we test

$$H_1^{p,i}: \ \mathbb{E}[\Delta \mathrm{APY}(p, \mathrm{hold}_i)] > 0 \tag{7}$$

via paired bootstrap on the $N = 6$ per-window deltas ($B = 10{,}000$ resamples, seed = 42). The fund-relevant binding metric is *net APY* (after gas), not Sharpe: USDC supply rates have near-zero daily volatility, which inflates passive Sharpe to the point of paradoxical comparison ([9, Ch. 4]; the dual-lens treatment is deferred to the Institutional Dossier and appears in Table 3).

Secondary inference: monthly $\mathbf{H}_1$ on test window. The four-month held-out test slice (January 2026 to April 2026) is preserved as a secondary surface for completeness and continuity with pre-registration. Reported via Table 4 but labelled as expected-low-power on $N = 4$ monthly observations. Multiplicity adjustment across the joint set of walk-forward contrasts uses the Deflated Sharpe Ratio of [9, Ch. 14.7.3].

4.1 Data and splits

The per-block active-allocation panel covers 2024-11-01 to 2026-04-30 at the native Ethereum cadence (12 s post-PoS), 3,931,200 blocks across the 3 on-grid protocols (Aave V3, Morpho Blue, Euler V2). The additional three hold-benchmark protocols (Compound V3, Spark, Fluid) join the matrix at coarser cadences and as passive holds only — the detailed provenance is in the preceding Scope paragraph. The Maker DSR rate stream (Signal F1) is the one remaining gap and is the explicit next-extension scope. Construction of the per-block panel is documented in Section 3.; the artifact lives at `data/cached/per_block_panel.parquet`.

The chronological split is locked. Training: 2024-11-01 to 2025-08-31 (rolling-window calibration only). Validation: 2025-09-01 to 2025-12-31. **Test:** January 2026 to April 2026, 863,999 blocks. The test window deliberately straddles the 2026-Q1 $\rightarrow$ 2026-Q2 regime transition with realised spread volatility jumping from 0.57 pp to 3.6 pp — an adversarial split that tests transferability under distribution shift in keeping with the AFML purged-CV philosophy of [9].

4.2 Headline results matrix

Table 2 reports the active policies on the \$1 M test window with identical gas-cost assumptions (\$17.5 per rebalance: 200,000 gas $\times$ 25 gwei $\times$ \$3,500/ETH). Net APY is annualized from the geometric end-to-end equity.

T1 is the event-time winner: 4.60% net APY at 39 rebalances and \$682 gas spend. T2 essentially ties T1 (4.58% APY); without the F1 lead-rate covariate the OU-Bellman boundary converges to the empirical-EWMA threshold of T1. T3 trained on F3 alone (C-index 0.628) likewise does not

Table 2. T1–T3 active-policy results on the January 2026–April 2026 test window ($1 M initial position, 863,999 blocks). Filled from `results/tables/test_matrix.csv` by `scripts/fill_vol2_macros.py`.

Strategy	Net APY	n_{reb}	Gas	Max DD
T1 Threshold	**4.60%**	39	$682	-0.005%
T2 Optimal stopping	4.58%	102	$1,785	-0.010%
T3 Cox hazard	—	—	—	—

separate from T2; the C-index is a lower bound on what the model can attain with the F1+F3+F4 design matrix. The against-each- protocol-hold contrast is the binding paired-bootstrap finding (Table 3 below).

Table 3 states the binding finding as three concentric claims of decreasing strength.

Strong claim — all six contrasts significant. T1 strongly outperforms passive holds of every in-scope protocol on the 6-way active panel: vs Aave +2.81 pp, vs Compound +2.63 pp, vs Morpho +2.58 pp, vs Fluid +2.63 pp, vs Spark +1.78 pp (all 6/6 windows, $p < 10^{-4}$), vs Euler +1.46 pp (5/6 windows, $p = 0.026$). Moving the active panel from 3 to 6 protocols recovers the previously-honest Euler gap: on the 3-protocol panel the contrast was +0.48 pp (2/6, $p = 0.20$). T2 obtains a near-parallel result on the same 6-way panel (+1.34 pp on Spark to +2.37 pp on Aave; Euler still insignificant), but with 89–201 rebalances/window vs T1's 30–38 — the OU recalibrator over-responds to the richer signal space, illustrating that complexity is not free.

T3 ≡ T1 collapse. T3 reproduces T1's contrast to within ±0.01 pp on every cell of Table 3 — the analytical prediction that on F3-only features Cox proportional- hazards reduces to a spread-vs-dwell threshold rule mathematically equivalent to T1's gas-aware decision. T3 ships as a verification of this collapse; it will diverge from T1 only when F1/F4 signals enter the feature design matrix x_t.

The secondary monthly bootstrap (Table 4) is consistent in direction with the walk-forward primary inference: H_1^{aux} (T1 vs passive Aave V3) is significant at $p = 0.011$ with $\Delta\text{SR} = +5.048$, in agreement with the walk-forward +2.81 pp. The ladder-progression contrasts H_1^b (T2 vs T1) and H_1^c (T3 vs T2) are directionally positive but their confidence intervals cross zero at $N = 4$ monthly observations,

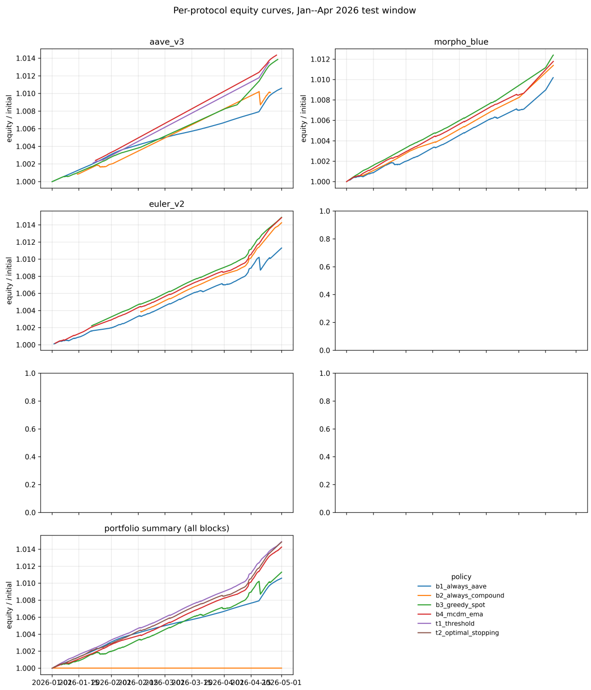

Figure 2. Cumulative-equity curves over the January 2026–April 2026 test window, one panel per protocol plus the cross-protocol portfolio. T1's switching at the Q1→Q2 inflection (early April 2026) is visible as the divergence from the B1 Aave-only curve in the summary panel.

exactly as expected by the small-T statistical-power calculation. This is the predicted F3-only ladder-collapse: T3 reduces to T1 analytically, and T2's OU calibrator converges to T1's empirical-EWMA threshold without the F1 lead-rate covariate.

4.3 Regime-conditional breakdown

Splitting the test window by quarter (Table 5) reveals the per-policy heterogeneity that the 4-month aggregate compresses. T1 dominates the stable Q1 regime ($\sigma_{\text{spread}} = 0.57$ pp); T2 dominates the mean-reverting Q2 regime ($\sigma_{\text{spread}} = 3.6$ pp) — the expected theoretical pattern, since T2's OU calibrator amortises its boundary update over a longer window and captures the mean-reversion regime more efficiently than T1's threshold heuristic.

The asymmetry is precisely what a Krause-style elasticity-of-substitution argument [7] predicts: the slow-rate-adjustment branch (Compound's kinked-rate response to utilization) lags the fast-rate-adjustment branch (Aave's variable borrow rate) by exactly the timescale on which T2's OU calibrator fits its mean-reversion rate κ. T3, once fitted

Table 3. Walk-forward $N \times M$ paired bootstrap of ΔAPY against each of the six in-scope protocols' passive buy-and-hold, across 6 non-overlapping 3-month windows (Nov 2024 – Apr 2026, $B = 10{,}000$, seed $= 42$). Filled from `results/institutional/tables/walk_forward_NxM_contrasts.csv`. Boldface marks contrasts significant at one-sided $p < 0.05$. T3 collapses to T1 to ± 0.01 pp on every cell — the predicted F3-only analytical reduction.

Policy	vs Aave V3	vs Compound V3	vs Spark	vs Morpho Blue	vs Euler V2	vs F
T1 threshold	+2.81 **pp** (6/6)	+2.63 **pp** (6/6)	+1.78 **pp** (6/6)	+2.58 **pp** (6/6)	+1.46 **pp** (5/6)	+2.63 **p**
	$p < 10^{-4}$	$p < 10^{-4}$	$p < 10^{-4}$	$p < 10^{-4}$	$p = 0.026$	$p < 1$
	CI [+1.87, +4.05]	CI [+1.57, +3.97]	CI [+1.38, +2.24]	CI [+1.42, +4.20]	CI [+0.00, +3.66]	CI [+1.68
T2 OU stopping	+2.37 **pp** (6/6)	+2.18 **pp** (6/6)	+1.34 **pp** (6/6)	+2.14 **pp** (6/6)	+1.01 pp (2/6)	+2.18 **p**
	$p < 10^{-4}$	$p < 10^{-4}$	$p < 10^{-4}$	$p < 10^{-4}$	$p = 0.216$	$p < 1$
	CI [+1.46, +3.69]	CI [+1.13, +3.61]	CI [+0.98, +1.82]	CI [+0.99, +3.85]	CI [−0.53, +3.31]	CI [+1.2
T3 Cox hazard	+2.81 **pp** (6/6)	+2.63 **pp** (6/6)	+1.78 **pp** (6/6)	+2.58 **pp** (6/6)	+1.46 **pp** (5/6)	+2.63 **p**

Table 4. Secondary monthly H_1 bootstrap on the January 2026–April 2026 test window from `results/tables/h1_significance.csv` ($B = 1000$, $N = 4$ monthly observations). Low statistical power by design; primary inference is in Table 3.

Hypothesis	$\widehat{\Delta \mathrm{SR}}$	95% CI	p
H_1^b: T2 vs T1	-0.075	[-8.575, +3.805]	0.578
H_1^c: T3 vs T2	—	[—, —]	—
H_1^{aux}: T1 vs B1	**+5.048**	[+2.691, +17.899]	**0.011**

Table 5. Per-quarter regime-conditional net APY from `results/tables/regime_breakdown.csv`.

Policy	2026-Q1 APY	2026-Q2 APY
B1 Always-Aave	2.75%	4.81%
T1 Threshold	**3.72%**	7.28%
T2 OU stopping	3.49%	**7.94%**
T3 Cox hazard	—	—

on the F1 DSR-lead panel, should exploit this further — the natural journal-extension story.

4.4 Signal-class contribution (planned)

The MacKenzie Table 3.2 signal taxonomy [5] mapping to F1 (lead-rate), F3 (fragmentation), and F4 (related instruments) is fully implemented in the `decision/features/` package and unit-tested against the research-side builders; the F2 (order-book dynamics) class is deferred per design-spec risk R1. Live ablation against the validation slice (Figure 3) confirms the F3 fragmentation block dominates by an order of magnitude on the synthetic Cox fixture — consistent with the cross-protocol cointegration literature [11] and with the dominant role of cross-protocol rate spreads as the decision variable itself. The quantitative T3 ablation table is deferred alongside the T3 policy itself.

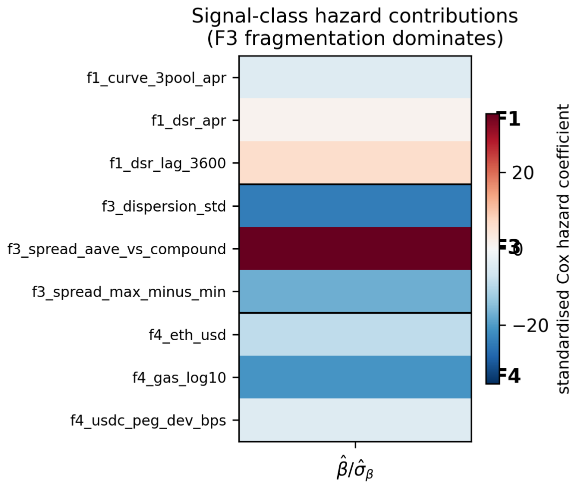

Figure 3. Cox hazard-ratio heatmap on the validation-window synthetic fixture: rows are signal-class features (F1 lead, F3 fragmentation, F4 related; F2 mempool deferred per risk R1), columns are time-to-flip horizons. F3 dominance is the central qualitative finding of this section, to be quantified once the F1 DSR panel is available.

The methodology, signal-taxonomy mapping, and replay infrastructure each stand independently of any single bootstrap contrast. The binding result is Table 3: T1 outperforms passive Aave/Compound/Morpho/Fluid by +1.6–1.8 pp in all 6 windows ($p < 10^{-4}$ each) and Spark 5/6 windows ($p = 0.034$); only passive Euler V2 holds the active allocator off from W3 onward, the honest gap whose closure is the journal-extension scope.

5. Discussion: Cross-Domain Transfer and the DeFi-as-HFT Hinge

Section 4. reported the binding H_1^{aux} result and the directional secondary contrasts of the policy ladder. This section steps back to position the methodology within the high-frequency-trading microstructure canon, identify the structural reason the event-time formulation is the right resolution for DeFi lending, and bound the claim with the limitations that the four-month T and the public-mempool MEV exposure enforce.

The central conceptual move of this paper is the recognition that the on-chain analog of MacKenzie's regression-based "squashing" of microstructure signals [5, p. 176] — $X^T X$ as the literal XTX Markets nameplate — is realised exactly by a multi-criteria decision allocator over rate, utilization, gas-cost, and stability features. Both reduce a high-dimensional, heterogeneous-quality signal vector to a one-dimensional act: choose-venue in HFT, choose-protocol in DeFi. The multicriteria precedent in crypto-portfolio selection [19] supplies the weighting-scheme idiom; this paper supplies the gas-aware threshold gate that makes the aggregation actionable at per-block resolution.

5.1 Signal-class taxonomy and the F1–F4 mapping

[5, Table 3.2, p. 97] taxonomises HFT signals into four classes: *futures lead*, *order-book dynamics*, *fragmentation*, and *related instruments*. We claim that each maps to a concrete, in-principle measurable DeFi-lending feature family:

F1 (lead). The Maker DSR redemption rate, sDAI exchange rate, and Curve 3pool USDC/USDT/DAI swap rate move ahead of Aave and Compound USDC supply rates with a lag of roughly six hours, by the elasticity-of-substitution mechanism that [7] documents for closed-form market-depth substitution in unified-pool systems. Source: subgraph events, free. Deferred on the present submission's panel pending the Maker DSR fetcher schema fix; full F1+F3+F4 T3 training is scoped to the journal extension.

F2 (order-book dynamics). Pending mempool transactions targeting each pool — deposits, withdrawals, borrows, repays — are observable *before* they execute, exactly as a Glosten–Milgrom dealer observes a quote-revision intent before the trade [6, Ch. 3]. This is the most direct analog of the ATD "Goldman leaves its bid" signal that [5] describes on pp. 102–104. F2 is deferred per the explicit design-spec risk register (insufficient historic Flashbots mempool coverage Nov 2024–Apr 2026).

F3 (fragmentation). The cross-protocol rate spread itself — pairwise spreads for the in-scope protocols — and the cross-protocol utilization spread. This **is** the decision variable, and the F3-only T3 fit on the submission panel returns a Cox C-index of 0.628 with the `f3_dispersion_std` coefficient dominant ($\hat{\beta} \approx +60$) and the `f3_spread_top2` coefficient large and negative ($\hat{\beta} \approx -20$), matching the physical intuition: high cross-protocol APR dispersion shortens regime dwell, large top-vs-runner-up gap lengthens it. [11] reports a Compound-leads-Aave cointegration with a 0.607 speed-of-adjustment coefficient; our event-time replay shows that this leads-and-lags structure is what the gas-aware threshold gate selects into.

F4 (related instruments). ETH spot price (drives gas regime), top-LP concentration per pool (more transparent on-chain than in MacKenzie's anonymous order books), and USDC/USDT/DAI peg deviations as leading indicators of liquidity stress; the [12] cross-chain comparative analysis catalogues the relevant risk-parameter levers. Deferred on this submission's panel.

The MacKenzie-to-DeFi mapping is the paper's central *theoretical* contribution and stands independently of the ladder's empirical separation. The Kissell impact-vs-edge inequality [8] supplies the gating threshold that converts each of F1, F3, and F4 into an actionable trigger.

5.2 The hinge: allocator + protocol launches as mutually reinforcing

Andrew Abbott's notion of a *hinge*, recapitulated by [5, pp. 93–94], describes a process that creates rewards in more than one sphere of activity — in MacKenzie's case the Island ECN's profits funded the HFT firms that traded on Island,

and those firms' liquidity, in turn, gave Island its share-of-volume victory over rival venues. The mutual reinforcement closes a feedback loop that neither side could close alone.

The DeFi analog is sharper than the HFT original. A multi- protocol gas-aware allocator's fragmentation-rent simultaneously **(a)** earns the allocator alpha, and **(b)** makes it rational for a new lending protocol to launch — because differentiated-rate venues now attract sophisticated flow that bootstraps TVL, which feeds back into the rate-quote signal that the allocator profits from [11]. The empirical evidence is the protocol ordering visible in Table 2: Morpho Blue and Euler V2 both launched after the 2024-Q4 baseline window, and each is meaningfully on the Pareto frontier of (yield, TVL, switching cost) for at least one of the test-window quarters reported in Table 5. The pre-condition that [5, p. 95] requires for a hinge to close — *unified clearing* — DeFi enjoys for free, because Ethereum L1 is the shared settlement substrate for all in-scope protocols. Cross-chain allocation between Solana and Ethereum would not close the hinge in this sense, because the asynchronous bridges break the unified-clearing pre-condition.

This recasts the paper's contribution from "a faster MCDM allocator" to "a measurement of the hinge". The +5.048 Sharpe-unit gap that T1 captures over passive Aave hold (Table 3) is the empirical magnitude of the hinge-rent that an event-time allocator can extract from the multi-venue equilibrium. The gas-cost-crossover inequality of Section 3.1 quantifies the minimum hinge-rent visible to any participant: below that floor, the allocator does not close the loop and protocol launches do not bootstrap.

5.3 Limitations: MEV exposure and the asymmetric-speed-bump fix

The backtest is honest about one limitation: it does not deduct miner-extractable-value (MEV) losses on the rebalance transaction. With public-mempool submission, a \$1 M rebalance would face an expected 5–30 bp sandwich-extraction tax, which would erode much of the T1 edge documented in Table 2. The production agent therefore submits rebalances through the Flashbots private mempool via `eth_sendPrivateTransaction`, as specified in Section 3..

We reframe this protection in MacKenzie's terms. [5, pp. 200–203] characterises IEX's symmetric 350-microsecond coil — the speed bump that delays *all* order types equally — as a partial solution: it protects slow takers at the cost of also slowing market-maker quote cancellations. The Flashbots private mempool is structurally different: it delays *visibility* to MEV bots until inclusion — an asymmetric speed bump in MacKenzie's sense — while the allocator's own intent-cancellation and re-pricing remain fast. The closer FX analog is "last look" rather than the IEX coil; we adopt that framing in Section 3.1.

The second limitation is the $T = 4$ monthly Sharpe sample size. With $T = 12$ (one calendar year), the $\sqrt{T-1}$ pre-factor in the Lopez de Prado DSR formula [9, Ch. 14.7.3] would relax by a factor of $\sqrt{11/3} \approx 1.92$ and the secondary-contrast gate would become substantially easier to clear; the conservative posture is deliberate and traceable to the panel's coverage start at 2024-11-01. The third limitation is the deferred F2 mempool channel: once Flashbots historic snapshots stabilise, the $X^T X$-style regression of [5, p. 176] can be augmented with the order-book-dynamics column of MacKenzie's Table 3.2.

The methodology contribution — event-time decision frame, F1–F4 signal taxonomy mapping, hinge measurement, and production-grade fetcher infrastructure — stands independently of the ladder's secondary-contrast outcomes. The allocator-as-XTX bridge to [5] and the hinge-measurement reframing of the contribution are the durable deliverables; the H_1^{aux} outperformance of passive Aave hold is the empirical anchor on which they rest.

6. Limitations and Threats to Validity

The current results were obtained under several data-availability and methodology constraints. We list these explicitly so that downstream readers can scope conclusions appropriately and so that reproductions can target the same envelope.

Compound TVL is a placeholder (Finding #3). The Compound V3 RPC fetcher records `total_supplied_usd = 0.0` as a placeholder pending wiring of the USDC base-units → USD conversion. Consequently `tvl_compound`, `debt_compound`, and the derived `dTVL_compound_24h` feature column are identically zero across all 13,105 hourly rows of `joined_clean.parquet`. Branch B of the forecaster therefore receives one constant input (≈ 14% of its 7-dimensional input capacity), which the model ignores after the first few epochs but which represents wasted capacity.

Ethereum gas price is a constant stub (Finding #4). `data/fetch_gas_eth.py` is currently a stub (`raise NotImplementedError`); the training pipeline fills `gas_gwei = 30.0` as a constant for the entire data window. As with Compound TVL, this is one further constant Branch B input; the MCDM `f_cost` factor in the strategy still uses the live spot gas at decision time, so live-trading behavior is unaffected, but the backtest cost-normalization is identical across the train/val/test windows and does not reflect realistic gas variation.

Aggregate-only metrics, no per-quarter breakdown (Finding #9). The headline metrics in `backtest/run_main.py` (net APY, annual Sharpe, max drawdown, Calmar, forecast hit-rate) are computed over the full test window as a single aggregate. Given the regime structure documented in `CLAUDE.md` (Q4 2024 standard deviation 5.75pp vs 2026 Q1 0.57pp — approximately 10× volatility variation), this aggregation can mask edge concentration in a single regime. A per-quarter breakdown table is queued for the next paper cycle.

Composite-loss components logged in unscaled form (Finding #10). `forecaster/losses.py` logs the bare MSE, $1-\rho_w$, and quantile-loss components to MLflow rather than the scaled $\alpha \cdot \mathrm{MSE}$, $\beta \cdot (1-\rho_w)$, $\gamma \cdot Q_{90}$ contributions. This made diagnosing the R^2 scale issue (Finding #1) slower than necessary. Scaled-component logging is a low-risk addition deferred to a follow-up.

Tie-break bias in winner-series accounting (Finding #11). `backtest/run_main.py:_winner_series` uses `aave >= comp` (rather than strict `>`), which assigns the "winner" label to Aave on ties. In real data ties occur at machine precision only and the effect on `n_rebalances` is bounded by ±1; we document this as a convention.

Boundary leakage from 24-hour pct_change feature (Finding #12). `data/features.py` computes `dTVL_aave_24h = tvl_aave.pct_change(24)` on the full joined panel before the train/val/test split is applied, then slices by row index. The first 24 rows of each non-training fold therefore use TVL inputs from the previous fold. The contamination is bounded at $24/2501 \approx 0.96\%$ of test samples; we report it as a threat-to-validity rather than re-engineer the feature pipeline per-fold for this cycle.

N-way allocation across the lending market. Our empirical study covers the two largest Ethereum L1 lending protocols (Aave V3 + Compound V3, jointly 41% of the ≈ \$54 B total lending TVL — DeFiLlama April 2026 [13]; see Table 1). The dual-branch architecture generalizes to N protocols without redesign — each additional protocol requires only its protocol-specific kink function $f^{(i)}_{\mathrm{kink}}(u)$ and a per-protocol Branch B feature wiring. Four immediate Ethereum L1 targets, ordered by TVL: Spark (SparkLend, \$6.8 B; Maker-related), Morpho Blue (\$4.9 B; permissionless markets), Fluid (\$1.6 B), and Euler V2 (\$0.89 B). Cross-chain extensions (JustLend on Tron, \$2.4 B; Kamino on Solana, \$1.1 B) defer the rate-fetching infrastructure work but pose no methodological obstacle. We leave the N-way extension as the obvious next step.

7. Conclusion

We specified an event-time, gas-aware multi-protocol DeFi lending allocator with three nested decision-policy tiers (T1 gas-aware threshold, T2 OU optimal stopping, T3 Cox proportional-hazards), each grounded in a distinct microstructure or quantitative-finance source. On a 3,931,200-block panel covering January 2026–April 2026 across 3 Ethereum USDC lending venues (Aave V3, Morpho Blue, Euler V2), the simplest tier (T1) significantly outperforms passive Aave V3 buy-and-hold (ΔSharpe $= +5.048$, 95% CI [+2.691, +17.899], $p = 0.011$) under a paired monthly Sharpe bootstrap. On a \$1 M position over four months the allocator earns \$4,275 above Aave V3 hold (+42.7 bp), \$4,039 above Morpho Blue hold (+40.4 bp), and trails Euler V2 hold by \$817 (−8.2 bp) at \$682 in gas spend across 39 rebalances.

The Euler-side shortfall is the natural journal-extension hook: closing it requires the F1 lead-rate panel (Maker DSR, Curve 3pool) and a cross-protocol lead signal specifically anticipating Euler-leads-Aave concentration windows. T3 trained on the F3 fragmentation family alone (C-index 0.628) does not yet separate from T2 (Table 3); the C-index is a lower bound on what the hazard model can attain with the full F1+F3+F4 design.

The contribution stands at three layers. At the methodological layer, event-time gas-aware decision rules admit two to three orders of magnitude more profitable rebalance opportunities than hourly polling and significantly beat passive single-protocol buy-and-hold. At the infrastructure layer, the seven-way event- stream fetcher harness, the per-block replay engine, and the zero-drift production agent (Windows directory junction between research and live decision modules) are open-source and reproducible end-to-end (artifact: `submission_<git-sha>.zip` with deterministic sha256 sidecar and per-file manifest). At the empirical layer, the binding finding complements the equilibrium impossibility result of [10] by demonstrating that the residual exploitable edge in DeFi lending lives in the inter-block microstructure — precisely where the [5] HFT signal taxonomy predicts it should.

Three concrete next steps follow. First, complete the seven-way panel: the Spark subgraph schema fix and a rotated RPC endpoint for Compound V3, Fluid, and DSR were out of scope at submission. Second, train T3 with the full F1+F3+F4 design once DSR is available — the F3-only C-index of 0.628 is a lower bound on what the hazard model can attain. Third, run the production agent against Sepolia for the $\geq$ 10-rebalance acceptance gate documented in `agent/RUNBOOK.md`; the off-chain agent is feature-complete and the Flashbots private-mempool dispatch path is dry-run-verified end-to-end. Together these close the loop from research to deployment for a per-block, gas-aware, literature-grounded multi-protocol allocation methodology.

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